COMP1511
Introduction to Discrete Mathematics

Lecture Notes
2023/24

The three main topics that are covered in this module are:

1. Combinatorics: Counting Methods
2. Discrete Probability Theory
3. Graph Theory

The lecture notes closely follow the material presented in Chapters 6, 7, 10 and 11 of the accompanying book (that is referred to in the rest of the text as *the book*):

> K. H. Rosen, **Discrete Mathematics and its Applications**, 7th Edition, McGraw Hill Education, 2013.

All of the material from the module syllabus will be covered in the lecture notes and presented in the lectures. The examination and coursework will only assess topics covered in the module syllabus. You should use the book as an additional source of problems to solve and as an alternative explanation of a topic. The book is there to supplement the lecture notes. Not all of the material from the chapters is covered in the lecture notes (i.e. lectures), and the lectures sometimes cover the material in a different order than the one in the book.

We expect you, as a student, to:

1. Attend lectures and carefully read the lecture notes.
2. Attend and engage in tutorials, completing all problems requested by your tutorial tutor.
3. Submit all courseworks.
4. To truly absorb the material covered, you should also practice exercises from the relevant sections of the book (the odd numbered ones have solutions in the back).

1 Combinatorics: Counting Methods (Corresponds to Sections 6.1 - 6.5 in the book)

Combinatorics is the study of arrangements of objects, and it is an important part of discrete mathematics. It was studied as long ago as the seventeenth century, when combinatorial questions arose in the study of gambling games. The counting of objects with certain properties is an important part of combinatorics. It is used, for example, in determining the complexity of algorithms, determining whether there are enough Internet protocol addressees to meet the demand, and when probabilities of events are computed.

1.1 Basic Counting Principles (Corresponds to Section 6.1 in the book)

The Multiplication Principle for Counting Procedures (or the Product Rule): If a procedure can be described as a sequence of m independent steps, with i_1 possible outcomes for the first step, i_2 possible outcomes for the second step, $\ldots$, and i_m possible outcomes for the m^{th} step, then the total number of composite outcomes of the procedure is $i_1 i_2 \cdots i_m$.

A *sequential counting procedure* is one that satisfies the following three properties:

1. The steps are *ordered*, i.e. any two sequences of different outcomes represents distinguishable composite outcomes.

2. The steps are *independent*, i.e. the number of outcomes of one step is not affected by the outcomes of the preceding steps.

3. The steps are *complete*, i.e. each composite outcome consists of a complete sequence of individual outcomes, one for each step.

Example

A binary sequence is a sequence whose elements come from the set $\{0,1\}$. For example, 100110 is a binary sequence of length 6.

(a) How many binary sequences of length 5 are there?

(b) How many binary sequences of length 5 begin with 101?

Example
How many ways are there to answer a true-false test with 12 questions.

(a) Every question must be answered.

(b) Any number of questions may be left unanswered.

Example

A licence plate consists of a sequence of 3 letters followed by 3 digits. How many licence plates are possible?

Example
(a) How many 7 letter strings are there?

(b) How many 7 letter palindromes are there? A palindrome is a string that reads the same forward and backward.

Example (k-element sequences)
Let $a_1, a_2, \ldots, a_k$ be a k-element sequence or a k-tuple (note that the order of the elements is important). Sometimes k-tuples are also denoted by $(a_1, a_2, \ldots, a_k)$. If there are n choices for each element of the sequence, then by the Multiplication Principle, the number of different k-element sequences is n^k.

Example
Find the number of distinct divisors of the integer 64800.

$64800 = 2^5 3^4 5^2$ prime factorization

Example
6 different books are arranged on a bookshelf.

(a) How many distinguishable orders of arrangements are there?

(b) Suppose that 2 of the 6 books are blue and should not be placed next to each other. How many different arrangements are there with 2 blue books not adjacent?

Example
Consider the problem of determining the number of outcomes when n dice are rolled.

(a) How many outcomes are possible when n dice are rolled if each die has a different color or is somehow different from the other?

(b) How many outcomes when 2 identical dice are rolled?

Example
What is the value of k after the following code, where $n_1, \ldots, n_m$ are positive integers, has been executed?

$k := 0$
for $i_1 := 1$ **to** n_1
    for $i_2 := 1$ **to** n_2
        .
        .
        .
      for $i_m := 1$ **to** n_m
        $k := k + 1$

The Addition Principle for Counting Procedures (or the Sum Rule): If a procedure can be broken up into m events or cases whose sets of outcomes are mutually exclusive, with j_1 possible outcomes for the first event, j_2 possible outcomes for the second event, ..., and j_m possible outcomes for the m^{th} event, then the total number of outcomes of the procedure is $j_1 + j_2 + \ldots + j_m$.

Example
On a bookshelf there are 5 different calculus books, 6 different linear algebra books and 7 different combinatorics books.

(a) In how many ways can a pair of books of different types be chosen?

(b) In how many ways can a single book of each type be chosen?

Example
How many integers are there from 1 through 9999 that have distinct digits?

Example

What is the value of k after the following code, where $n_1, \ldots, n_m$ are positive integers, has been executed?

$k := 0$
for $i_1 := 1$ **to** n_1
    $k := k + 1$
for $i_2 := 1$ **to** n_2
    $k := k + 1$
        .
        .
        .
for $i_m := 1$ **to** n_m
    $k := k + 1$

Counting What You Do Not Want

Example

How many binary sequences of length 5 have at least two 1's?

Example
In a standard deck of cards there are 4 suits: clubs, diamonds, hearts and spades. Each suit contains 13 cards: $2, 3, \ldots, 10, J, Q, K, A$. So there are $4 \cdot 13 = 52$ cards in the deck.
Suppose we choose a sequence of 4 cards from the deck, with replacement, i.e. one card at a time is chosen and it is placed back into the deck before the next choice.

(a) In how many ways can a sequence of 4 cards be chosen, with replacement, so that some card is repeated?

(b) In how many ways can a sequence of 4 cards be chosen, with replacement, so that the fourth card is the first repeated card?

(c) In how many ways can a sequence of 4 cards be chosen, with replacement, so that the fourth card is a repeat, but not necessarily the first repeat?

Principle of Inclusion-Exclusion: Let A_1 and A_2 be two sets. The number of ways to select an element from A_1 is $|A_1|$, the number of ways to select an element from A_2 is $|A_2|$, and the number of ways to select an element from A_1 or A_2 is $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$.

Example

How many binary sequences of length 6 start with 1 **or** end with 00?

1.2 Selections (Corresponds to Sections 6.3 and 6.5 in the book)

Suppose that a child is offered a bag containing three types of sweets, aniseed drops (A), butter mints (B) and cherry drops (C). In how many ways can the child select two sweets?

This problem is not defined well, since we are not told whether or not the same type of sweet can be selected twice (i.e. is AA allowed?), and does it matter in which order the sweets are selected (i.e. is AB to be treated as a different selection from BA, or not?).

There are 4 cases to consider:

1. Repeats allowed and order matters. This gives 9 possibilities:
   AA, AB, AC, BA, BB, BC, CA, CB, CC.
2. Repeats allowed and order does not matter. This gives 6 possibilities:
   AA, AB, AC, BB, BC, CC.
3. Repeats not allowed and order matters. This gives 6 possibilities:
   AB, AC, BA, BC, CA, CB.
4. Repeats not allowed and order does not matter. This gives 3 possibilities:
   AB, AC, BC.

1.2.1 Permutations

Consider the selection of k objects from a set of n objects where **order matters and repetition is not allowed**. Any such selection is called a *k-permutation*. The total number of k-permutations of a set of n (distinct) objects is denoted by $P(n,k)$.

Example
$X = \{a,b,c,d\}$
2-permutations are:

Factorial notation: for a non-negative integer n, n *factorial* is defined as

$$n! = \begin{cases} 1 & \text{if } n = 0 \\ n(n-1)(n-2)\ldots 1 & \text{if } n \geq 1. \end{cases}$$

Note that $P(n,n) = n!$.

Theorem 1.1 *For integers k and n such that* $0 \leq k \leq n$*, the number of k-permutations of a set of n objects is* $P(n,k) = \frac{n!}{(n-k)!}$.

Example

How many 7 letter strings with no repeated letters can be formed from English alphabet?

Example

In how many ways can 5 books be chosen from a pile of 10 distinct books and arranged on a bookshelf?

Example

A computer center has to run 9 different programs in a row, 4 of them in FORTRAN and 5 in pascal. Find the number of possible schedules for running the programs if fortran programs must be first, followed by pascal programs.

1.2.2 Combinations

Consider the selection of k objects from a set of n objects where **order does not matter and repetition is not allowed**. Any such selection is called a k-*combination*. The total number of k-combinations of a set of n (distinct) objects is denoted by $C(n,k)$ or $\binom{n}{k}$ ("n choose k").

Example
$X = \{a,b,c,d\}$
2-combinations are:

Theorem 1.2 *For integers k and n such that $0 \leq k \leq n$, the total number of k-combinations of a set of n objects is $C(n,k) = \frac{n!}{k!(n-k)!}$.*

Note that by Theorem **??**, $C(n,k) = C(n, n-k)$.

Example

(a) How many subsets of three integers can be chosen from the set $\{1,2,\ldots,10\}$?

(b) How many subsets in (a) contain 3?

(c) How many of the subsets in (a) contain at least one multiple of 3?

Example (counting subsets of a finite set)
Number of ways to choose a subset from a set X of n objects:

(a) Any subset.

(b) Any nonempty subset.

(c) Any k-element subset.

Example

A new club starts with n different members. If each member shakes hands with everyone else exactly once, how many handshakes take place?

Example
Suppose a juggling club has 17 members.

(a) In how many ways can they choose 4 different members to be president, vice president, secretary and treasurer?

(b) In how many ways can they choose 4 different members to form a juggling team?

(c) In how many ways can a captain and 3 regular members be chosen?

Example
How many 4-digit integers are there where each digit is greater than those to the right?

1.2.3 Permutations with Unlimited Repetition

Consider the selection of k objects from a set of n objects where **order matters and repetition is allowed**. Any such selection is called a *k-permutation with unlimited repetition* of a set of n objects.

Theorem 1.3 *For positive integers k and n, the number of k-permutations with unlimited repetition of a set of n objects is n^k.*

Example
How many 5 letter strings can be formed from English alphabet?

Example
Let $A = \{1, \ldots, 6\}$ and $B = \{a, b, c, d, e\}$.

(a) Find the number of functions from A to B.

(b) Find the number of one-to-one functions from B to A.

(c) Find the number of onto functions from A to B.

1.2.4 Combinations with Unlimited Repetition

Consider the selection of k objects from a set of n objects where **order does not matter and repetition is allowed**. Any such selection is called a *k-combination with unlimited repetition* of a set of n objects.

We will show the following:
The number of k-combinations with unlimited repetition of a set of n objects
$=$ the number of nonnegative integer solutions of $x_1 + x_2 + \ldots + x_n = k$
$=$ the number of distributions of k identical balls into n distinct boxes
$=$ the number of sequences of length $k+n-1$ consisting of k 0's and $n-1$ $|$'s.

Let A be the set of all k-combinations with unlimited repetition of a set X with n elements.
Let B be the set of all nonnegative integer solutions of $x_1 + x_2 + \ldots + x_n = k$.
Let C be the set of all distributions of k identical balls into n distinct boxes.
Let D be the set of all sequences of length $k+n-1$ consisting of k 0's and $n-1$ $|$'s.

So we want to show that $|A| = |B| = |C| = |D|$.

• We first show that $|A| = |B|$.

Let $X = \{1, 2, \ldots, n\}$.

Suppose that we select
x_1 objects of type 1,
x_2 objects of type 2,
$\ldots$ and
x_n objects of type n.

Then, for an n-tuple of integers $(x_1, x_2, \ldots, x_n)$ the following holds:

for every $i \in \{1, \ldots, n\}$, $x_i \geq 0$, and $x_1 + x_2 + \ldots + x_n = k$.

Each n-tuple of nonnegative integers that is a solution of the linear equation $x_1 + x_2 + \ldots + x_n = k$ corresponds to exactly one k-combination with unlimited repetition and vice versa.

So there is a one-to-one correspondence between A and B.

$\Rightarrow |A| = |B|$

- Next we show that $|B| = |C|$.

Example

In how many ways can 7 identical balls be distributed into 3 distinct boxes?

• In general, $|C| = |D|$.

Theorem 1.4 *For positive integers k and n, the number of k-combinations with unlimited repetition of a set of n objects is* $C(k+n-1,k) = \frac{(k+n-1)!}{k!(n-1)!}$.

Example
Find the number of nonnegative integer solutions of $x_1 + x_2 + \ldots + x_5 = 30$.

Example
Find the number of positive integer solutions of $x_1 + x_2 + \ldots + x_5 = 30$.

Example
A bookshelf holds 11 different books in a row. In how many ways can we choose 4 books so that no 2 consecutive books are chosen?

	Ordered selections	Unordered selections
No repetition	$P(n,k)$	$C(n,k)$
Unlimited repetition allowed	n^k	$C(k+n-1,k)$

1.3 Binomial Coefficients and Combinatorial Identities (Corresponds to Section 6.4 in the book)

At first glance the expression $(a+b)^n$ does not have much to do with combinations, but it turns out that we can obtain the formula for the expansion of $(a+b)^n$ by using the formula for k-combinations of n objects.

$(a+b)^2 = (a+b)(a+b) = aa+ab+ba+bb = a^2+2ab+b^2$

$(a+b)^3 = (a+b)(a+b)(a+b) = aaa+aab+aba+abb+baa+bab+bba+bbb = a^3+3a^2b+3ab^2+b^3$

Theorem 1.5 Binomial Theorem
If a and b are real numbers and n is a positive integer, then $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$.

The numbers $\binom{n}{k}$ are known as *binomial coefficients* because they appear in the expansion of $(a+b)^n$.

Example
What is the coefficient of $x^{12}y^{13}$ in the expansion of $(x+y)^{25}$?

Example
Expand $(3x-2y)^4$ using the Binomial Theorem.

Example
Find the coefficient of a^4b^5 in the expansion of $(2a-b)^9$.

Example
Find the coefficient of $x^2y^3z^4$ in the expansion of $(x+y+z)^9$.

A **combinatorial proof** of an identity is a proof that uses counting arguments to prove that both sides of the identity count the same objects but in different ways or a proof that is based on showing that there is a bijection between the sets of objects counted by the two sides of the identity.

Example
Prove that for a positive integer n, $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.

Proof using Binomial Theorem:

Combinatorial proof:

Theorem 1.6 Pascal's Identity
Let n and k be positive integers with $n \geq k$. Then $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$.

Remark: It is also possible to prove Pascal's Identity by algebraic manipulation from the formula for $\binom{n}{k}$.

Theorem 1.7 Vandermonde's Identity
Let m,n and r be nonnegative integers with r not exceeding either m or n. Then $\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{r-k}\binom{n}{k}$.

1.4 The Pigeonhole Principle (Corresponds to Section 6.2 in the book)

The *Pigeonhole Principle* (also known as the *Dirichlet Principle*) is sometimes useful in answering the question: Is there an item having a given property?

Pigeonhole Principle: If n pigeons fly into k pigeonholes and $k < n$, then some pigeonhole contains at least two pigeons.

Example
Ten persons have first name Alice, Bernard and Charles, and last names Lee, McDuff and Ng. Show that at least two persons have the same first and last names.

Generalized Pigeonhole Principle: If n pigeons fly into k pigeonholes, then some pigeonhole contains at least $\lceil \frac{n}{k} \rceil$ pigeons.

Example
Among 100 people there are at least $\lceil \frac{100}{12} \rceil = 9$ who were born in the same month.

Example
What is the minimum number of students required in a discrete mathematics class to be sure that at least 6 will receive the same grade, if there are 5 possible grades, A,B,C,D and F?